Approximate Positions of the Planets

Introduction

Lower accuracy formulae for planetary positions have a number of important applications when one doesn't need the full accuracy of an integrated ephemeris. They are often used in observation scheduling, telescope pointing, and prediction of certain phenomena as well as in the planning and design of spacecraft missions.

Approximate positions of the planets may be found by using Keplerian formulae with their associated elements and rates. Such elements are not intended to represent any sort of mean; they are simply the result of being adjusted for a best fit. As such, it must be noted that the elements are not valid outside the given time-interval over which they were fit.

High precision ephemerides for the planets are available via the Horizons system (/horizons/).

Accuracy

The table below lists nominal errors in heliocentric longitude, λ, latitude, φ, and distance, ρ, using this approximation of planetary positions.

	1800 AD — 2050 AD			3000 BC — 3000 AD		
	λ (arcsec)	**ϕ (arcsec)**	**ρ (1000 km)**	**λ (arcsec)**	**ϕ (arcsec)**	**ρ (1000 km)**
Mercury	15	1	1	20	15	1
Venus	20	1	4	40	30	8
EM Bary†	20	8	6	40	15	15
Mars	40	2	25	100	40	30
Jupiter	400	10	600	600	100	1000
Saturn	600	25	1500	1000	100	4000
Uranus	50	2	1000	2000	30	8000

	1800 AD — 2050 AD			3000 BC — 3000 AD		
	λ (arcsec)	**ϕ (arcsec)**	**ρ (1000 km)**	**λ (arcsec)**	**ϕ (arcsec)**	**ρ (1000 km)**
Neptune	10	1	200	400	15	4000

† **EM Bary** = Earth/Moon Barycenter

Formulae for using the Keplerian elements

Keplerian elements given in the tables below are

- $a_o, \dot{a}$ semi-major axis [au, au/century]
- $e_o, \dot{e}$ eccentricity
- $I_o, \dot{I}$ inclination [degrees, degrees/century]
- $L_o, \dot{L}$ mean longitude [degrees, degrees/century]
- $\varpi_o, \dot{\varpi}$longitude of perihelion [degrees, degrees/century]
- $\Omega_o, \dot{\Omega}$ longitude of the ascending node [degrees, degrees/century]

In order to obtain the coordinates of one of the planets at a given Julian ephemeris date, $\mathrm{T_{eph}}$ (equivalent to the post-2006 IAU Barycentric Dynamical Timescale, JDTDB),

1. Compute the value of each of that planet's six elements: $a = a_o + \dot{a}\mathrm{T}$, etc., where T, the number of centuries past J2000.0, is T = $(\mathrm{T_{eph}} - 2451545.0)/36525$.

2. Compute the argument of perihelion, ω, and the mean anomaly, M:

$$\omega = \varpi - \Omega \ ; \ M = L \ - \ \varpi \ + \ b\mathrm{T}^2 \ + \ c\cos(f\mathrm{T}) \ + \ s\sin(f\mathrm{T})$$

3. Adjust the mean anomaly M to its equivalent angle in the range $-180^{\mathrm{o}} \leq M \leq +180^{\mathrm{o}}$ and then obtain the eccentric anomaly, E, from the solution of Kepler's equation (see below):

$$M \ = \ E - e^* \sin E$$

where $e^* \ = \ 180/\pi \ e \ = \ 57.29578 \ e$.

4. Compute the planet's heliocentric coordinates in its orbital plane, $\mathbf{r}'$, with the x'-axis aligned from the focus to the perihelion:

$$x' = a(\cos E - e) \quad ; \quad y' = a\sqrt{1-e^2} \ \sin E \quad ; \quad z' = 0.$$

5. Compute the coordinates, $\mathbf{r}_{ecl}$, in the J2000 ecliptic plane, with the x-axis aligned toward the equinox considering the frame rotation:

$$\mathbf{r}_{ecl} \ = \mathcal{M}\mathbf{r}' \ \equiv \ \mathcal{R}_z(-\Omega)\mathcal{R}_x(-I)\mathcal{R}_z(-\omega)\mathbf{r}'$$

applied using the corresponding vector rotation such that

$$\begin{aligned}
x_{ecl} &= (\cos\omega\cos\Omega - \sin\omega\sin\Omega\cos I)\ x' + (-\sin\omega\cos\Omega - \cos\omega\sin\Omega\cos I)\ y' \\
y_{ecl} &= (\cos\omega\sin\Omega + \sin\omega\cos\Omega\cos I)\ x' + (-\sin\omega\sin\Omega + \cos\omega\cos\Omega\cos I)\ y' \\
z_{ecl} &= (\sin\omega\sin I)\ x' + (\cos\omega\sin I)\ y'
\end{aligned}$$

6. If desired, obtain the equatorial coordinates in the "ICRF" or "J2000 frame", $\mathbf{r_{eq}}$:

$$\begin{aligned}
x_{eq} &= x_{ecl} \\
y_{eq} &= +\cos\varepsilon\ y_{ecl} - \sin\varepsilon\ z_{ecl} \\
z_{eq} &= +\sin\varepsilon\ y_{ecl} + \cos\varepsilon\ z_{ecl}
\end{aligned}$$

where the obliquity at J2000 is $\varepsilon = 23\overset{\circ}{.}43928$.

Solution of Kepler's Equation

$$M = E - e^* \sin E$$

Given the mean anomaly, M, and the eccentricity, e^*, both in degrees, start with[††]

$$E_0 = M + e^* \sin M$$

and iterate the following three equations, with $n = 0, 1, 2, \ldots$, until $|\Delta E| \le tol$:

$$\Delta M = M - (E_n - e^* \sin E_n)\ ;\ \ \Delta E = \Delta M/(1 - e\cos E_n)\ ;\ \ E_{n+1} = E_n + \Delta E.$$

For the approximate formulae in this present context, $tol = 10^{-6}$ degrees is sufficient.

[††] This starting guess ensures faster convergence, but $E_0 = M$ or $E_0 = M - e^* \sin M$ could also be used.

Keplerian Elements and Rates

Table 1

Keplerian elements and their rates, with respect to the mean ecliptic and equinox of J2000, valid for the time-interval 1800 AD - 2050 AD.

```
           a            e            I            L          long.peri.     long.node.
        au, au/Cy    rad, rad/Cy    deg, deg/Cy    deg, deg/Cy    deg, deg/Cy    deg, deg/Cy
-------------------------------------------------------------------------------------------------------
Mercury  0.38709927    0.20563593    7.00497902    252.25032350   77.45779628   48.3307
       0.00000037    0.00001906   -0.00594749  149472.67411175    0.16047689   -0.125340
Venus    0.72333566    0.00677672    3.39467605    181.97909950  131.60246718   76.6798
       0.00000390   -0.00004107   -0.00078890   58517.81538729    0.00268329   -0.2776941
EM Bary  1.00000261    0.01671123   -0.00001531    100.46457166  102.93768193    0.0
       0.00000562   -0.00004392   -0.01294668   35999.37244981    0.32327364    0.0
Mars     1.52371034    0.09339410    1.84969142     -4.55343205  -23.94362959   49.559538
       0.00001847    0.00007882   -0.00813131   19140.30268499    0.44441088   -0.2925734
Jupiter  5.20288700    0.04838624    1.30439695     34.39644051   14.72847983  100.47390
      -0.00011607   -0.00013253   -0.00183714    3034.74612775    0.21252668    0.2046910
Saturn   9.53667594    0.05386179    2.48599187     49.95424423   92.59887831  113.6624
      -0.00125060   -0.00050991    0.00193609    1222.49362201   -0.41897216   -0.2886779
Uranus  19.18916464    0.04725744    0.77263783    313.23810451  170.95427630   74.016
      -0.00196176   -0.00004397   -0.00242939     428.48202785    0.40805281    0.04240589
Neptune 30.06992276    0.00859048    1.77004347    -55.12002969   44.96476227  131.784
       0.00026291    0.00005105    0.00035372    218.45945325   -0.32241464   -0.00508664
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EM Bary = Earth/Moon Barycenter
```

Table 2a

Keplerian elements and their rates, with respect to the mean ecliptic and equinox of J2000, valid for the time-interval 3000 BC -- 3000 AD. NOTE: the computation of M for Jupiter through Neptune *must* be augmented by the additional terms given in Table 2b (below).

```
            a            e            I            L          long.peri.     long.node.
         au, au/Cy    rad, rad/Cy    deg, deg/Cy    deg, deg/Cy    deg, deg/Cy    deg, deg/Cy
-----------------------------------------------------------------------------------------------
Mercury   0.38709843    0.20563661    7.00559432    252.25166724    77.45771895    48.3396
      0.00000000    0.00002123   -0.00590158  149472.67486623    0.15940013   -0.122141
Venus     0.72332102    0.00676399    3.39777545    181.97970850   131.76755713    76.6726
     -0.00000026   -0.00005107    0.00043494   58517.81560260    0.05679648   -0.2727417
EM Bary   1.00000018    0.01673163   -0.00054346    100.46691572   102.93005885   -5.112
     -0.00000003   -0.00003661   -0.01337178   35999.37306329    0.31795260   -0.241238
Mars      1.52371243    0.09336511    1.85181869    -4.56813164   -23.91744784    49.713209
      0.00000097    0.00009149   -0.00724757   19140.29934243    0.45223625   -0.2685243
Jupiter   5.20248019    0.04853590    1.29861416    34.33479152    14.27495244   100.29282
     -0.00002864    0.00018026   -0.00322699    3034.90371757    0.18199196    0.1302461
Saturn    9.54149883    0.05550825    2.49424102    50.07571329    92.86136063   113.63998
     -0.00003065   -0.00032044    0.00451969    1222.11494724    0.54179478   -0.2501500
Uranus   19.18797948    0.04685740    0.77298127    314.20276625   172.43404441    73.962
     -0.00020455   -0.00001550   -0.00180155     428.49512595    0.09266985    0.05739699
Neptune  30.06952752    0.00895439    1.77005520    304.22289287    46.68158724   131.78
      0.00006447    0.00000818    0.00022400    218.46515314    0.01009938   -0.00606302
-----------------------------------------------------------------------------------------------
EM Bary = Earth/Moon Barycenter
```

Table 2b

Additional terms which must be added to the computation of M for Jupiter through Neptune, 3000 BC to 3000 AD, as described in the related document.

```
            b            c            s            f
-------------------------------------------------------------
Jupiter   -0.00012452   0.06064060   -0.35635438   38.35125000
Saturn     0.00025899  -0.13434469    0.87320147   38.35125000
Uranus     0.00058331  -0.97731848    0.17689245    7.67025000
Neptune   -0.00041348   0.68346318   -0.10162547    7.67025000
-------------------------------------------------------------
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Reference

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